

6032C Electrification of Residential Buildings

POLY PMNA - Polytechnic Study Hub

Semester 6 | Program Elective | Credits 4 | Periods 60

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Core areas: residential wiring safety, wiring systems, protection devices, lighting calculation, sub-circuits, earthing, certification, star labelling and estimation.

Course Overview and Roadmap

Objectives: design internal wiring of residential buildings, draw wiring and schematic diagrams, and recognize the importance of safety and protection devices.

CO1: Outline specifications and uses of components for residential electrification.

CO2: Choose lighting scheme and calculate lighting points. CO3: Identify estimate for suitable earthing as per IS standards. CO4: Prepare material schedule and estimate for residential electrification.

Roadmap: M1 safety, wiring systems, accessories and cable selection. M2 illumination, lamps, lighting schemes and lighting point calculation. M3 symbols, sub-circuits and earthing estimate. M4 material schedule, certification, star labelling and estimation.

Module Chapters

Module 1: Electric shock depends on current, path, body resistance and contact time. Residential wiring needs correct insulation, protection, earthing and safe accident procedure. Wiring systems include conduit wiring, accessories, main switches, MCB, MCCB, RCCB, ELCB, switches, sockets, ceiling roses, lamp holders, fuses and MDB.

Module 2: Illumination terms include luminous flux, illumination, luminous efficiency and coefficient of utilization. Lamps are compared by lumen per watt, lifespan and CRI. Lighting systems include direct, semi-direct, diffused and indirect schemes.

Module 3: Standard symbols, conductor colour scheme, labelling, sub-circuit calculation, connected load, cable size, protective device rating and pipe/plate earthing estimate are required. Earthing provides low resistance path for fault current and shock protection.

Module 4: Estimation includes standard format, location, material cost, tax, contingency, labour cost and validity period. Certifications include BIS, IEC, CE and UL. Residential estimate includes layout, wiring diagram, schedule of materials, labour and testing.

Formula Bank, Practice and Answers

Formula bank: Power = voltage x current for simple single phase estimate. Energy = kW x time. Illumination = luminous flux / area. Lighting watts = area x watts per square metre. Estimate amount = quantity x rate.

Solved example: 24 sq.m room, 8 W/sq.m, 12 W LED lamps. Required watts = $24 \times 8 = 192$ W. Number of lamps = $192 / 12 = 16$ lamps.

Expected questions: explain shock precautions, conduit wiring, wiring accessories, cable selection, lighting schemes, sub-circuit design, pipe earthing, star labelling and material estimate preparation.

Model answer guide: include drawing, formula, specifications, unit, quantity, rate, labour, tax, contingency, safety devices and earthing notes. Do not ignore protection rating.